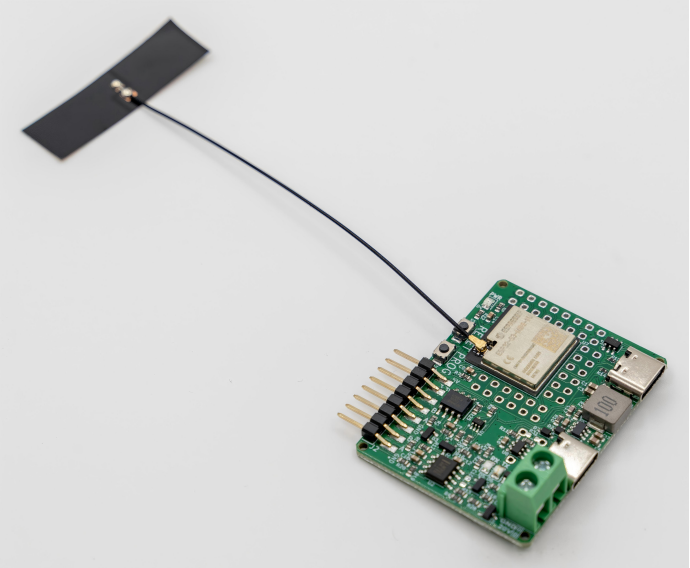
PROGRAMMABLE MOTION CONTROL

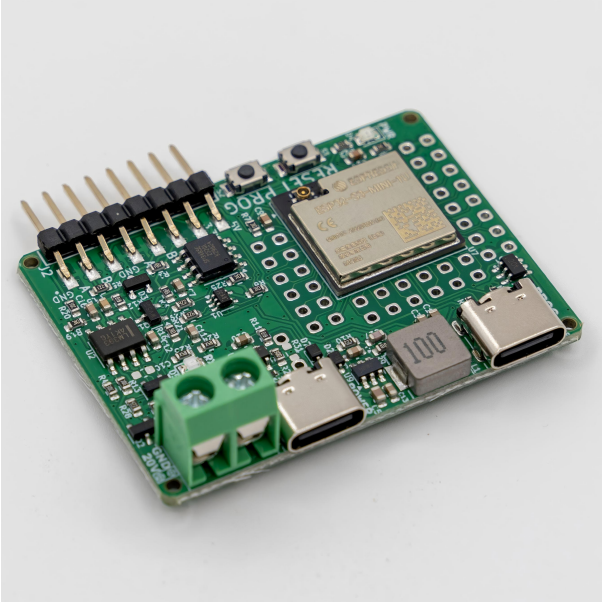
RS485 Controller

ESP32-S3 development board for Gearotons servomotors and other RS485 devices

Affordable and Simple All-in-One Motion Control

From Education to Innovation





Actual product photographs. External antenna included; cables, power supply and motors are separate system components.

11.7 g
Including attached antenna

230400 baud
Canonical example setting

PCB 0.5
Hardware covered here

A compact controller for shared-bus motion

Run motor-control logic directly on the ESP32-S3. The board combines native USB programming, a half-duplex RS485 interface, a USB-C PD power input and accessible GPIO pads in a compact open PCB.

Embedded control: dual-core LX7 processor, up to 240 MHz; 2.4 GHz Wi-Fi and Bluetooth 5 LE capabilities depend on application firmware.

Simple wiring: two parallel A/B/GND groups, automatic RS485 direction circuitry, reset and programming buttons.

Motor-power integration: standard assembly requests 20 V USB-C PD, with a direct motor-power output at the screw terminal.

SPECIFICATIONS

Electrical & power

Design values are distinguished from measurements and qualified product limits.

Parameter	Specification / conditions
Processor / module	ESP32-S3-MINI-1U; dual-core Xtensa LX7, up to 240 MHz.
Wireless hardware	2.4 GHz Wi-Fi 802.11b/g/n and Bluetooth 5 LE; external antenna included. Wireless control requires suitable firmware.
Programming supply, J5	5 V nominal USB input. Native ESP32-S3 USB data connection; motor power is supplied separately.
PD power input, J3	Standard design requests 20 V using CH221K and R27 = 200 kΩ. Actual voltage depends on the source and successful PD negotiation.
Motor-power output, J1	Pin 1 = J3 VBUS rail (labelled +20 V); pin 2 = GND. Direct pass-through, not a separately regulated 20 V output.
Logic / RS485 interface	3.3 V logic; SP3485EN half-duplex differential transceiver. Controller, USB and motor grounds are common; no galvanic isolation.
RS485 example setting	230400 baud, 8 data bits, no parity, 1 stop bit. TX = GPIO4; RX = GPIO5. Board maximum baud rate is not specified.
Bus loading / protection	R5 = 220 Ω across A/B. R1 = 560 Ω pulls A high; R3 = 560 Ω pulls B low. SM712 transient suppressor on the bus.
Auxiliary power pins, J2	3.3 V rail and J5-associated 5 V rail. Available external current is not specified. The 5 V rail is not generated from the PD input.

Verify motor power before connecting

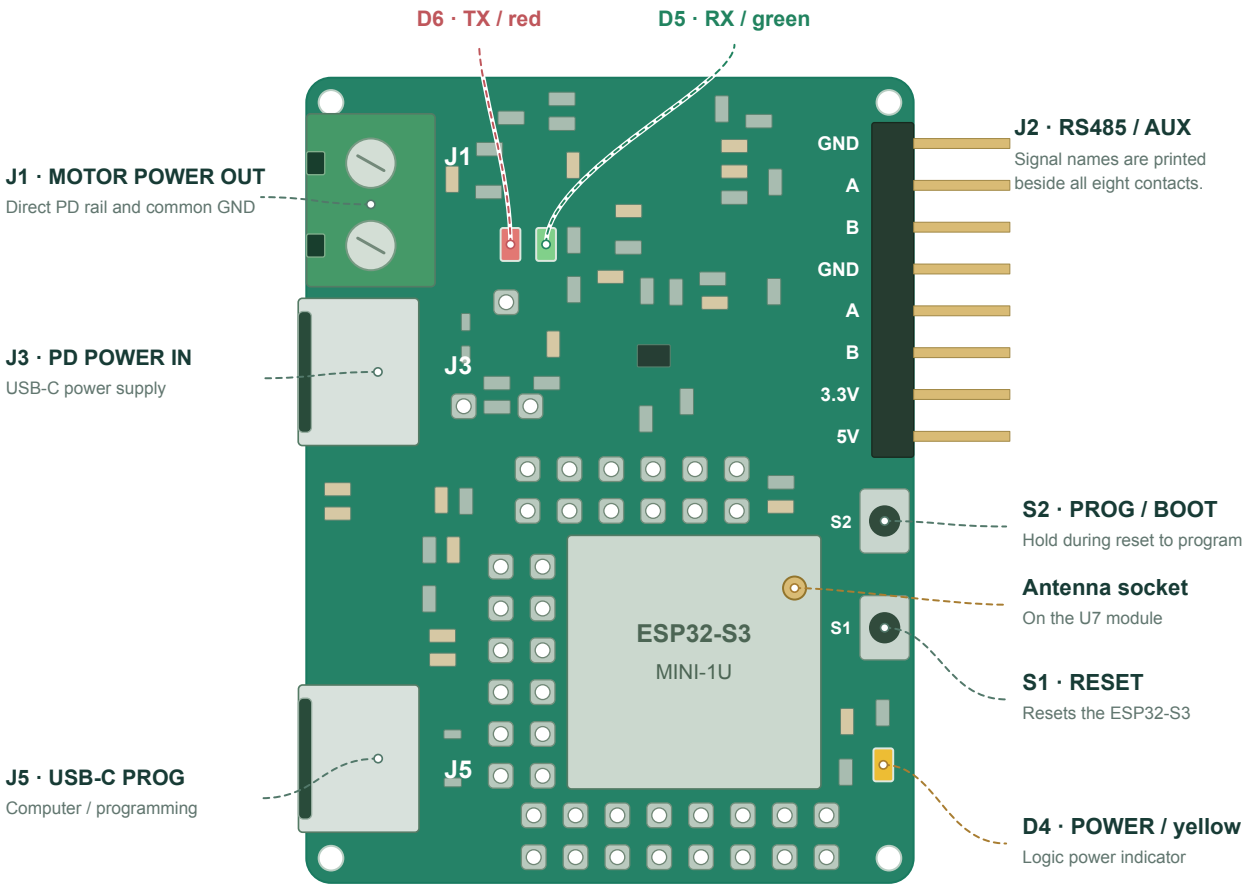
Use a USB-C PD supply that explicitly offers 20 V and a suitable cable. Measure J1 before attaching motors; a non-PD or unsuccessful source may leave the rail at a lower voltage. J5 alone does not provide motor power at J1.

Not yet specified: board current consumption, continuous motor-power pass-through current, maximum cable length and environmental limits. Component ratings are not board ratings; product-level ESD/EMC ratings have not been established.

INTERFACES · PHYSICAL LOCATOR

Power, programming & status

Locate connectors, switches and indicator LEDs before connecting or programming the controller.



U7 · ESP32-S3-MINI-1U

COMPONENT SIDE (B.Cu) · USB sockets at left · J2 header at right

Component-side view (B.Cu). Curved dashed leaders identify the connectors, switches, LEDs and U7 antenna socket. The eight J2 contacts are labelled by signal. Attach the included antenna before using the radio; the electrical reference follows.

Separate the two USB-C functions

J3: USB-C PD motor-power input. **J5:** native USB programming and 5 V logic power. The J3 USB data pins are unconnected. USB-C connector shape does not indicate which function it provides.

INTERFACES · ELECTRICAL REFERENCE

Connector pinout & controls

The two RS485 groups on J2 are electrically parallel; all grounds share one reference.

J2 pin	Signal	Use
1	GND	Common ground
2	A	RS485 A
3	B	RS485 B
4	GND	Same net as pin 1
5	A	Same bus as pin 2
6	B	Same bus as pin 3
7	3.3 V	Logic supply rail; external load not rated
8	5 V	Programming USB supply rail; external load not rated

Power, programming, switches and indicators

Reference	Function
J1 · POWER OUT	Two-way screw terminal. Pin 1 is the positive PD rail; pin 2 is GND. Verify polarity on the board.
J3 · PD POWER IN	USB-C power input and PD negotiation. This socket does not program the ESP32-S3.
J5 · USB-C PROG	Native USB D–/D+ connect to GPIO19/GPIO20. Use a data-capable USB cable.
S1 · RESET	Pulls the module EN signal low to reset the controller.
S2 · PROG	Pulls GPIO0 low; hold during reset to enter ROM download mode.
D4 · yellow LED	Logic-power indication.
D5 · green LED	Receive activity; indication depends on bus traffic.
D6 · red LED	Transmit activity; indication depends on bus traffic.

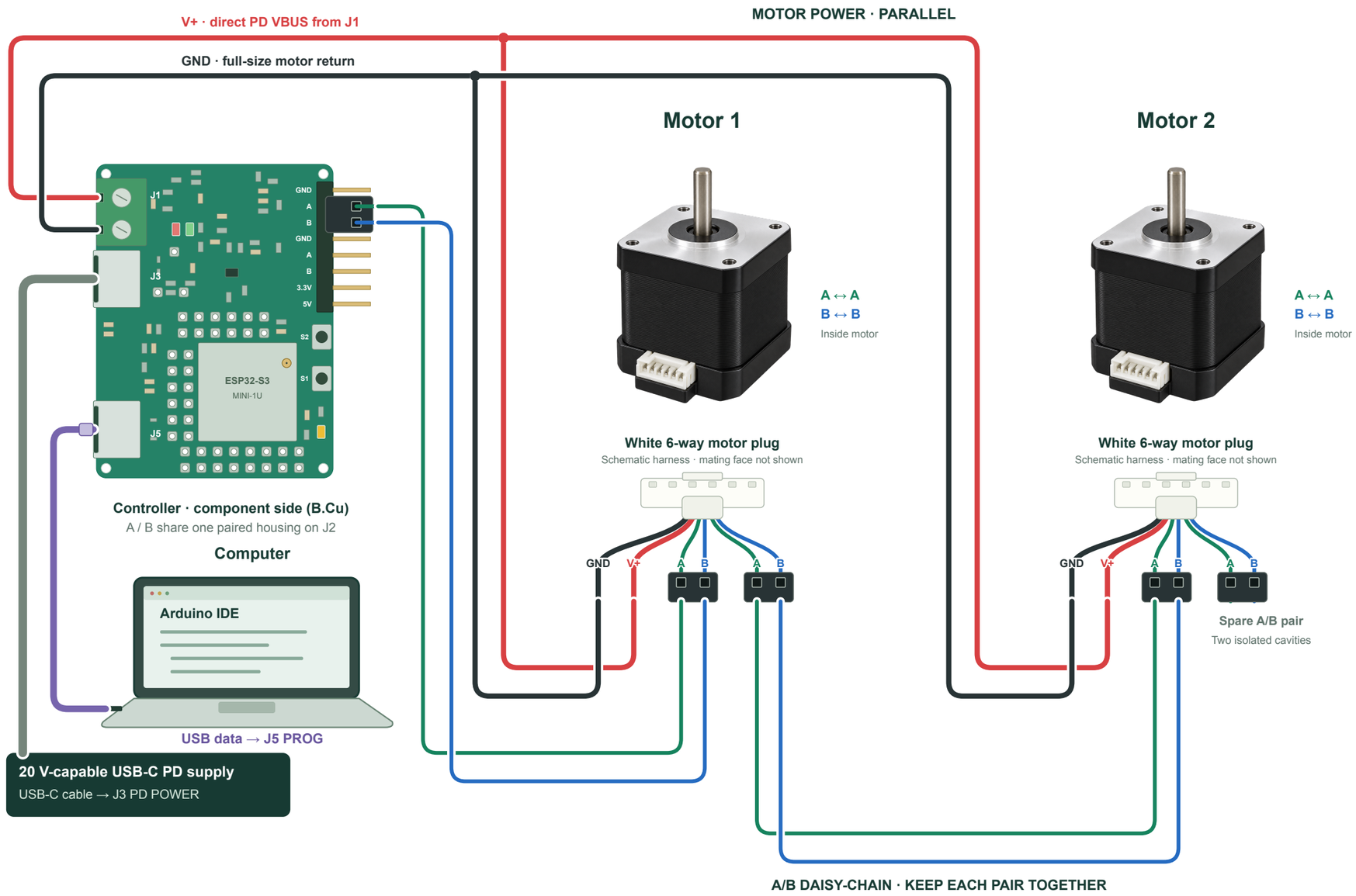
Keep motor supply voltage off J2. Its 3.3 V and 5 V pins are auxiliary logic rails, not motor-power terminals. Refer to the physical locator and actual PCB labels for orientation.

Natural illustration · computer USB to J5 PROG, PD supply to J3 · electrical reference follows.

GEAROTONS / RS485 CONTROLLER

Two motors. One complete connection map.

USB-C PD input · parallel motor power · one continuous RS485 bus



Power in parallel. A/B continues through Motor 1 to Motor 2.

J1 and J2 GND are common on the PCB; both motors have full-size power-return wires.
20 V is nominal after PD negotiation. Wire with power off; verify voltage and polarity.
A/B lines show electrical pairing; select compatible mating parts. Each black shell has two isolated cavities.

Computer: J5 USB-C PROG. Motor-power supply: J3 PD POWER IN. Connector mating faces are not shown; match actual labels.

SYSTEM WIRING

Connect two servomotors

One 20 V-capable USB-C PD source can feed J3 and supply the motors through J1.

Use the following net-by-net connections for both motors. Connect the data bus along one route and distribute motor supply/return wiring for the combined motor current.

Net	Controller connection	Motor 1 and Motor 2
+20 V	J1 pin 1	Each motor V+ / positive supply
GND	J1 pin 2 and J2 pin 1 or 4	Each motor GND / supply return
A	J2 pin 2 or 5	A to A, onward along the bus
B	J2 pin 3 or 6	B to B, onward along the bus

1. Check power while disconnected. With motors disconnected, connect a 20 V-capable PD source to J3 and measure J1. Check polarity, then turn the source off before wiring the motors.

2. Wire the common ground and motor supply. Connect both motor supply returns and controller GND. Motor returns must be sized for motor current; the data-ground lead must not be the only motor-current return.

3. Connect the differential bus. Connect A to A and B to B. Route A/B as a twisted pair along one bus, keeping branches short. Give each motor a distinct alias before addressed motion commands.

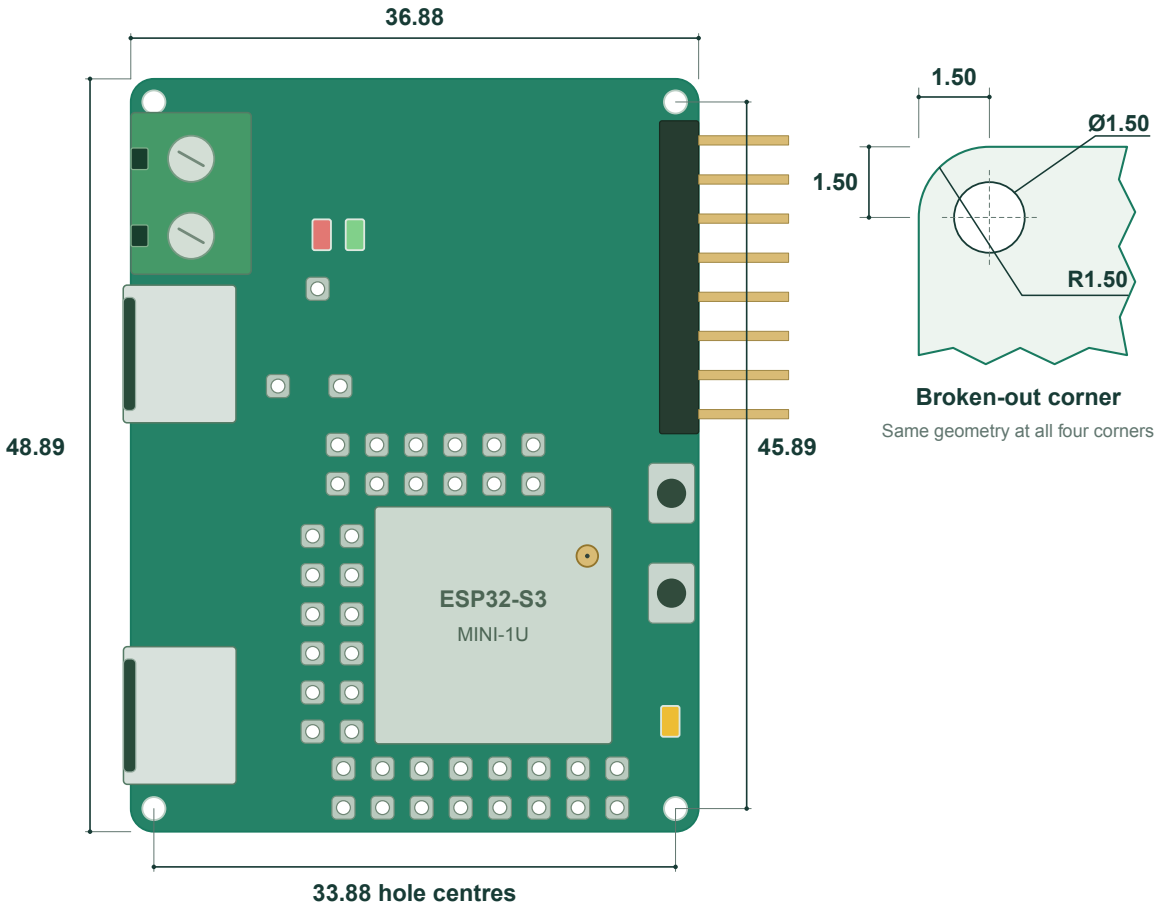
4. Power and test deliberately. Secure both motors before discovery or movement examples. The power supply, cable and distribution wiring must support the combined load.

The M17 motor reference specifies 12–24 V operation; the controller standard PD request is 20 V. R5 already loads A/B with 220 Ω. Select additional end termination for the complete bus instead of adding resistors at every motor. No fixed motor-count guarantee is specified.

MECHANICAL INTEGRATION

Detailed PCB dimensions

Nominal revision 0.5 KiCad geometry. Dimensions are in millimetres and exclude connector overhang and antenna.



mm · 4 × Ø1.50 mounting holes · centres 1.50 from adjacent edges · corners R1.50

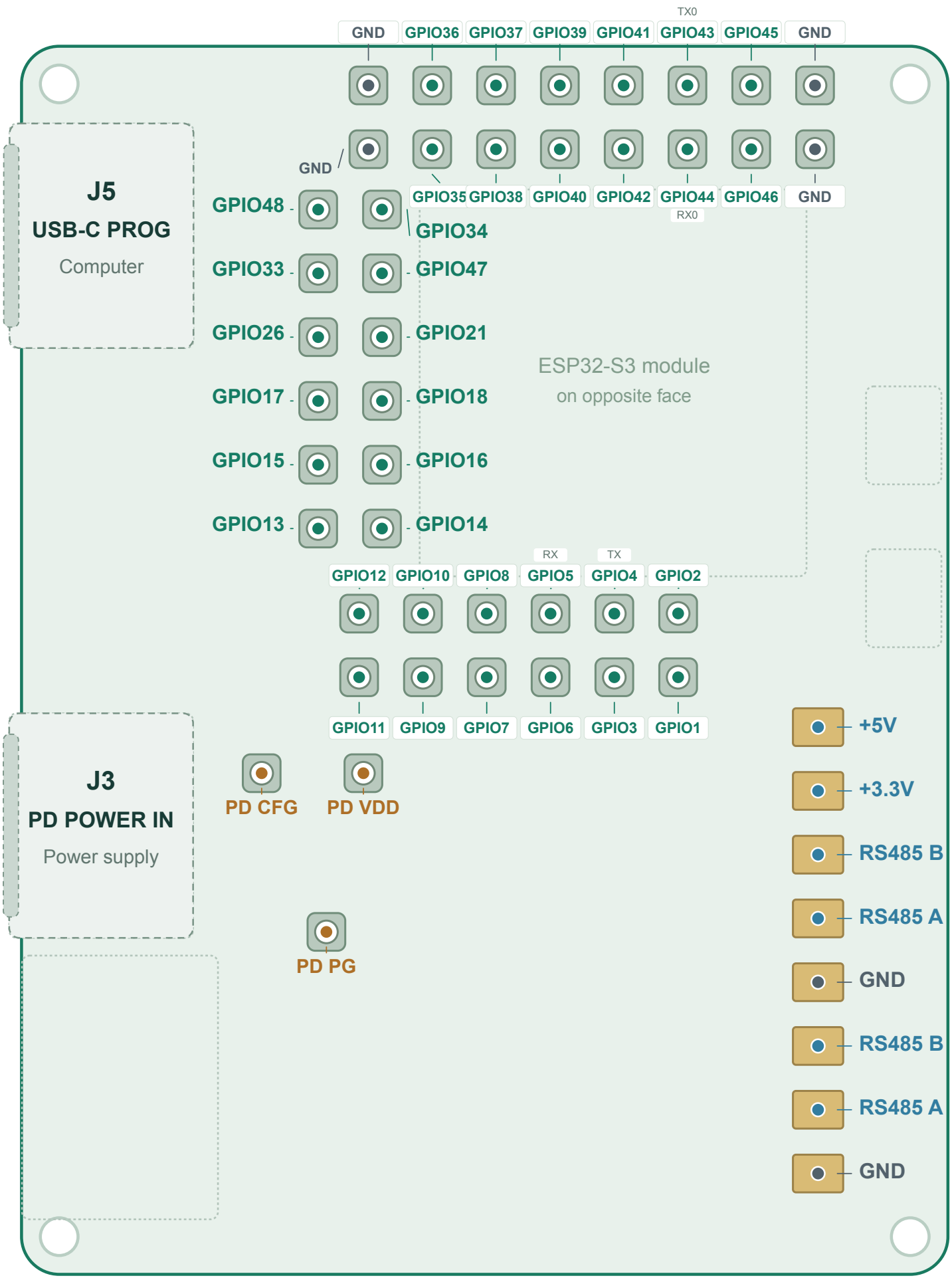
Use the dimensioned drawing for board outline, corner radius, mounting-hole diameter, hole-centre spacing and edge insets. These are design values without manufacturing tolerances; confirm enclosure clearances on a physical unit.

Mounting and assembled envelope

PCB thickness is 1.6 mm nominal. Keep clearance around underside components and connector solder joints. Overall assembled height and antenna/cable envelope are not specified. The Ø1.50 mm holes are not M2 or M3 clearance holes.

Breakout headers — solder required

Rear / label-side view (F.Cu). Breakout headers are not fitted; solder headers or wires to the required through-hole pads.



REAR / LABEL SIDE (F.Cu) · looking toward the PCB

USB-C sockets and module are on the opposite face; dashed outlines locate them.

Round holes: solder headers or wires here. Gold rectangular pads: surface service points.

Every pad is labelled by signal at its physical location. Green: GPIO. Gray: GND. Amber: PD-control points. Blue labels: surface service pads. Dashed socket and module outlines locate components fitted on the opposite face.

Shared functions: GPIO4 / TX and GPIO5 / RX are connected to onboard RS485. GPIO43 / TX0 and GPIO44 / RX0 are UART0. GPIO0/3/45/46 are module strapping signals; GPIO0 is connected to PROG. Use 3.3 V logic and avoid disturbing boot, USB or RS485.

PD CFG, PD VDD and PD PG are not GPIO. The gold rectangular service pads are surface pads, not through holes. Breakout holes are unpopulated; the fitted J2 RS485 header is separate and appears on the component-side locator.

Module reference: Official Espressif ESP32-S3-MINI-1 / MINI-1U datasheet (PDF). Zoom in on this vector drawing to inspect individual labels.

BRING-UP

Getting started

Prepare the hardware and software, establish communication, then enable motion deliberately.

1. **Prepare the hardware.** Attach the included external antenna. With power off, check the motor polarity and A/B/GND connections. Secure the mechanism before running any movement example.

2. **Use the programming port.** Connect J5 USB-C PROG with a data-capable cable. J3 is the separate USB-C PD power input, not a programming port.

3. **Install Arduino support.** Install the Arduino ESP32 board package and “Servomotor by Gearotons” from Library Manager. Select an ESP32-S3 target; configure native USB CDC for the USB serial console as appropriate for the board package.

4. **Enter download mode if needed.** Hold S2 PROG, press and release S1 RESET, then release PROG. Select the detected USB device and upload the application.

5. **Verify motor power and identity.** Check J1 voltage before connecting motors. Use the correct motor alias and ensure the mechanism can safely perform one full rotation before running the Arduino example.

Arduino example for the controller

The following installation guide uses the Arduino IDE instructions and screenshot from the M17 servomotor datasheet. The Arduino basic motor-control program is retained unchanged from M17 datasheet v1.10; setup notes remain outside the code.

The example commands motor motion

The program enables the motor drive, requests one shaft rotation in one second, then disables the drive. Match the motor alias and serial connection to the intended motor, and secure the mechanism before running it.

GETTING STARTED · ARDUINO IDE

Install the Arduino library

Arduino Library Manager instructions and original screenshot from the M17 servomotor datasheet.

Install the library through the Arduino Library Manager:

1. Open Arduino IDE
2. Go to Sketch → Include Library → Manage Libraries
3. Search for "Servomotor"
4. Find the library called "Servomotor by Gearotons" and click Install



Original M17 screenshot; Tools → Manage Libraries is the route shown. Choose the Gearotons library and the latest version. The pictured ESP32C3 target is illustrative; select the ESP32-S3 target for this controller.

GETTING STARTED · CANONICAL ARDUINO

Arduino: basic motor control

Verbatim program from the English M17 servomotor datasheet, v1.10 · reference PDF page 13.

Install “Servomotor by Gearotons” using Arduino Library Manager. The unchanged ESP32 pin definitions in this reference are GPIO4 TX and GPIO5 RX.

```
// Minimal Arduino example: Trapezoid move using built-in unit conversions
// Goal: spin the motor exactly 1 rotation in 1 second, then stop.
// Sequence:
//  enable MOSFETs -> trapezoidMove(1.0 rotations, 1.0 seconds) -> wait 1.1s -> disable MOSFETs.
//
// Notes:
// - This uses the library's unit conversion (no raw counts/timesteps).
// - Configure Serial1 pins for your board (ESP32 example pins below).
// - Motor is created AFTER Serial1.begin(...) so hardware UART pins are set first.

#include <Servomotor.h>

#define ALIAS 'X'                   // Device alias
#define BAUD 230400                 // RS485 UART baud rate
#define DISPLACEMENT_ROTATIONS 1.0f // 1 rotation
#define DURATION_SECONDS 1.0f       // 1 second
#define TOLERANCE_PERCENT 10        // +10% wait margin because the motor's clock is not
                                    //  perfectly accurate
#define WAIT_MS ((unsigned long)(DURATION_SECONDS * 1000.0f * (100 + TOLERANCE_PERCENT) / 100))

// Example RS485 pins for ESP32 DevKit (change as needed for your board)
#if defined(ESP32)
#define RS485_TXD 4             // TX pin to RS485 transceiver
#define RS485_RXD 5             // RX pin from RS485 transceiver
#endif

void setup() {
  Serial.begin(115200); // Console serial for debugging

  // Create the motor; serial port opens on first instantiation.
#if defined(ESP32)
  Servomotor motor(ALIAS, Serial1, RS485_RXD, RS485_TXD);
#else
  Servomotor motor(ALIAS, Serial1);
#endif

  // Use units: rotations for position, seconds for time
  motor.setPositionUnit(PositionUnit::SHAFT_ROTATIONS);
  motor.setTimeUnit(TimeUnit::SECONDS);

  motor.enableMosfets();
  motor.trapezoidMove(DISPLACEMENT_ROTATIONS, DURATION_SECONDS);
  delay(WAIT_MS);
  motor.disableMosfets();
}

void loop() {
}
```

This is the program printed in M17 datasheet v1.10, not a newer working-tree variant. Source snapshot and PDF correspondence are recorded in the accompanying example-provenance JSON.

DESIGN REFERENCE

Controller schematic & resources

The appendix reproduces the full controller sheet with ESP32-S3-MINI-1U retained as one module block.

Full controller schematic

The following landscape appendix embeds the supplied revision 0.5 controller schematic as vector content. It includes the board power, PD negotiation, native USB, RS485, controls and indicator circuitry. Zoom in to read component values without a raster-image resolution limit.

Source sheet: `PCB/0.5/schematic/0.5-index-schTop.pdf`. The module symbol and its external connections remain intact; no ESP32 module-internal schematic has been added.

Official module documentation

Espressif ESP32-S3-MINI-1 / ESP32-S3-MINI-1U datasheet
Manufacturer reference for the module pin definitions, boot configuration, radio, electrical specifications and module mechanical details.

Software, tutorials and design files

Controller source: github.com/tomrodinger/RS485_Controller
Motor setup and protocol: tutorial.gearotons.com
Arduino library: github.com/tomrodinger/Servomotor_Arduino_Library
Feedback: tutorial.gearotons.com/feedback

Specification and review basis

PCB 0.5 schematic, board geometry and assembly BOM; supplied firmware examples; owner-provided 11.7 g mass and included-antenna statement; M17 English datasheet v1.10 (Arduino program) and its Arduino IDE installation guide; official Espressif, MaxLinear and TI component references. No component current, environmental or ESD rating has been promoted to a board-level guarantee.

DS v0.4 · owner review draft

Prepared 3 November 2026. Physical product measurements and limits remain subject to confirmation where marked not specified. Hardware/software source is supplied under the repository MIT licence. Earlier review PDFs are preserved separately.

Full controller schematic

PCB 0.5 · module retained as a black box · A3 landscape appendix for comfortable digital zoom and larger-format printing.

